Ishan Sehgal

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Profile

Aspiring engineer,Currently a final-year Electronics and Communication student at Guru Nanak Dev University, Amritsar. Skilled in designing autonomous systems, with practical experience in simulation, control, and implementation of reliable, real-world solutions.

Skills

Programming Languages: C++, Python;**Robotics Frameworks:** ROS, ROS 2, Open-RMF, MoveIt2; **Hardware:**Raspberry Pi, YD LiDAR, ESP32,Depth cameras,GPS,3D-Lidars,Robotics Manipulators,Motor Encoders;, **Software Tools:** Fusion 360, Gazebo, SLAM, Nav/Nav2, Behavior Trees, Linux

Experience

Onsite Internship at Rigbetel Labs LLP *January 2025 - June 2025*

- Conducted hands-on projects with real robots in an onsite environment, gaining practical experience in robotics implementation.
- Successfully converted two AGVs (Automated Guided Vehicles) to fully autonomous AMRs (Autonomous Mobile Robots).
- Executed multiple robotics projects involving real-world applications and autonomous navigation systems.

EYSIP 2024 Summer Internship at e-Yantra *May 2024 - July 2024*

- Developed behavior trees for autonomous eBot and UR5 arm, improving reliability by reducing manual resets and downtime.
- Designed fail-safe mechanisms and implemented auto-localization, enabling Nav2 stack initialization from any location.

E-YRC 2023 at IIT Bombay *September 2023 - January 2024*

- Advanced to Stage 2, securing AIR-13 in the Cosmo Logistics theme.
- Developed an autonomous sorting and packaging system using a robotic arm and mobile robot, optimizing algorithms.

Mentorship at Rigbetel Labs *June 2023 - September 2023*

- Received comprehensive ROS training, designed robot components in Fusion 360, and simulated robot behavior in Gazebo.
- Gained proficiency in ROS architecture and robot simulation, enabling efficient prototyping of new designs.

Projects

Autonomous eBot Project

- Developed an autonomous eBot using ROS 2 Foxy, integrating Ylidar, motor encoders, and an ASUS i5 motherboard.
- Implemented SLAM for navigation and obstacle avoidance, achieving consistent localization accuracy across dynamic environments.
- GitHub link: E-bot

Autonomous Robocar Project

- Integrated Arduino UNO, ultrasonic sensors, and Raspberry Pi Zero to develop an obstacle avoidance car with remote control capabilities.

Awards & Achievements

- Final stage qualifier of EYRC-2023 (top 15 teams).
- Secured first position in the 'I am Engineer' contest at university..
- Secured first prize in Science Exhibition TechNit'16.
- Member of Technical Club at Guru Nanak Dev University under ESF.

Education

Guru Nanak Dev University Amritsar *2021-2025 (ongoing)*

- B.Tech Electronics and Communication
- CGPA: 7.23

Secondary Education

- MGN Public School Jalandhar *2018-2021*
- 12^{th} Percentage: 85.8%